
Advanced Draw And Understand: Free-Shape Visual Prompting for Pixel-Level Multimodal Comprehension

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Abstract

Advanced Draw And Understand (ADNU) elevates multimodal large language models to genuine pixel-level comprehension by replacing the conventional rectangular proxy with learnable free-shape visual prompts. A plugin visual prompt encoder converts arbitrary user sketches into compact token sequences, while a dynamic gating mechanism lets the LLM decide which prompts to attend to, eliminating performance inversion when multiple regions are marked. Extensive experiments on MDVP-Bench and five downstream tasks show that ADNU improves Referral accuracy by 4.7 pp, Reasoning accuracy by 5.6 pp, and zero-shot Chinese OCR by 24.7 pp over the strongest bounding-box baseline, yet requires no extra detection labels during pre-training. Code and multilingual data will be made publicly available.

1. Introduction

Multimodal Large Language Models (MLLMs) have rapidly evolved, showing strong performance in vision-language tasks (Achiam et al., 2023; Liu et al., 2023a; Li et al., 2023a; Alayrac et al., 2022; Zhu et al., 2023a; Dai et al., 2023; Huang et al., 2023; Peng et al., 2023; Wu et al., 2024; Wang et al., 2023a; Chen et al., 2023a; Sun et al., 2024; Ye et al., 2023; 2024). By aligning visual encoders (Radford et al., 2021; Dosovitskiy et al., 2021) with LLMs (Touvron et al., 2023; Bai et al., 2023;

Young et al., 2024; Lu et al., 2024; Chen et al., 2024; Wang et al., 2023b; Liu et al., 2024), these models can perceive and reason about images and even support embodied interaction (Driess et al., 2023). Recent surveys provide a comprehensive overview of this rapidly evolving landscape (Yin et al., 2024; Huang et al., 2024). However, the way users interact with them is still quite limited. Most models only accept coarse prompts like bounding boxes or points (Kirillov et al., 2023; Zhang et al., 2023; Ke et al., 2023). While boxes work well for standard rectangular objects, they are merely rough approximations. When dealing with irregular shapes or complex geometries, a box inevitably includes irrelevant background noise, confusing the model.

1.1. Related Work

The limitation of coarse-grained interaction becomes critical when precision matters. If a user wants to highlight a specific part of an object or a winding path, a bounding box is too crude. Recent works like LISA (Lai et al., 2023), SEEM (Zou et al., 2023), Semantic-SAM (Li et al., 2023b), Shikra (Chen et al., 2023b), PixelLM (Ren et al., 2023), and GLaMM (Rasheed et al., 2024) have attempted to integrate segmentation capabilities to provide more fine-grained understanding. However, these methods often rely on detection-based pre-training or expensive segmentation labels, which limits their scalability and data efficiency. Other approaches like Visual Prompting (Bahng et al., 2022), Painter (Wang et al., 2023c), and SegGPT (Wang et al., 2023d) explore in-context learning but struggle with pixel-level precision for arbitrary shapes. In parallel, vision-LLMs with unified pixel-level interfaces and detection-oriented reasoning (Fei et al., 2024; Pi et al., 2023) further highlight the need for fine-grained grounding within general-purpose MLLMs. Beyond static images, video-centric multimodal models (Munasinghe et al., 2023; Fei et al., 2023; Zhu et al., 2023b; Yu et al., 2023a) explore pixel-level reasoning and planning over

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spatio-temporal sequences.

Furthermore, current methods often fail when handling multiple prompts simultaneously—a phenomenon known as "performance inversion" (Liu et al., 2023b; Chen et al., 2022; Sofiuk et al., 2022). In interactive segmentation, adding more points or boxes paradoxically lowers accuracy. This suggests that existing models lack an effective mechanism to prioritize and attend to multiple regions of interest, leading to confusion when the scene becomes complex.

1.2. Our Approach

To address these issues, we propose Advanced Draw And Understand (ADNU), a framework designed for genuine pixel-level comprehension. Instead of forcing users to use rigid boxes, ADNU allows for free-shape visual prompts—scribbles, polygons, or arbitrary sketches—mirroring how humans naturally highlight information.

Our approach introduces two key components to make this work. First, a Plug-in Visual Prompt Encoder converts these arbitrary user sketches into compact token sequences, capturing precise geometric details without background interference. Second, we incorporate a Dynamic Gating Mechanism that acts as an adaptive filter. It allows the model to dynamically weigh the importance of different prompts, effectively solving the performance inversion problem by focusing only on what matters.

Crucially, ADNU is data-efficient. Unlike many prior works that rely on expensive detection or segmentation labels for pre-training, our method requires no such supervision. Experiments on MDVP-Bench (Shi et al., 2024) and downstream tasks show that ADNU not only supports more natural interaction but also significantly outperforms strong bounding-box baselines.

In short, our contributions are:

- We introduce ADNU, enabling MLLMs to understand pixel-level free-shape prompts, moving beyond the constraints of bounding boxes.
- We propose a specialized encoder and a dynamic gating mechanism to robustly handle arbitrary sketches and multi-target queries, resolving performance inversion under dense prompts.
- We construct the large-scale Chinese-MDVP benchmark and a MAE-style pre-training pipeline, and show that our full system delivers substantial practical gains on MDVP-Bench and Chinese-MDVP without relying on additional detection su-

pervision during pre-training.

2. Dataset

To evaluate the pixel-level comprehension capabilities of MLLMs, we utilize the MDVP-Bench (Shi et al., 2024), a comprehensive benchmark specifically designed for Multimodal Fine-grained Visual Prompting. MDVP-Bench comprises high-quality images with precise pixel-level annotations, covering diverse scenarios such as natural scenes, document understanding, and medical imaging.

However, a critical limitation of the original MDVP-Instruct-Data is its linguistic distribution: approximately 95% of the data is in English. This imbalance creates a significant barrier for evaluating and training models in non-English contexts, leading to high failure rates in few-shot or zero-shot scenarios for other languages, particularly Chinese.

2.1. Multilingual MDVP (Chinese-MDVP)

To address this "Multilingual Cultural Bias," we introduce an extended dataset, Chinese-MDVP, which serves as a crucial contribution to the original benchmark. We constructed this dataset using a scalable automated pipeline:

- **Data Generation Pipeline:** We leveraged the Segment Anything Model (SAM) (Kirillov et al., 2023) to extract fine-grained object masks and utilized BLIP-2 (Li et al., 2023a) to generate rich textual descriptions for these regions.
- **Scale and Diversity:** The resulting dataset contains 200,000 region descriptions and 400,000 Question-Answer (QA) pairs in Chinese.
- **Cultural Alignment:** A key feature of Chinese-MDVP is the inclusion of culturally specific entities (e.g., "Zongzi" for the Dragon Boat Festival). We injected cultural priors into the system prompts during generation, which improved the recall of low-resource entities by 12%.

This multilingual expansion not only tests the model's cross-lingual visual grounding abilities but also mitigates the performance degradation typically observed when transferring English-centric visual encoders to other languages. It is complementary to large-scale generic image-text corpora and detection datasets (Schuhmann et al., 2022; Kuznetsova et al., 2020; Shao et al., 2019; Lin et al., 2014; Ordonez et al., 2011), which primarily focus on English annotations

and coarse region labels. Complementary evaluation benchmarks for MLLMs and LVLs (Fu et al., 2023; Xu et al., 2023; Liu et al., 2023c; Yu et al., 2023b; Ying et al., 2024; Cui et al., 2024; Liu et al., 2022) further motivate our choice of tasks and metrics.

3. Methodology

3.1. Overview

We introduce the ADNU framework, which addresses the limitations of coarse-grained interactions and performance inversion in MLLMs. Our approach consists of five key innovations designed to enhance prompt flexibility, adaptability, and scalability. The inference process is detailed in Algorithm 1.

The workflow begins with processing free-shape visual prompts (Section 3.2), which are then filtered by a Dynamic Gating mechanism (Section 3.3) to suppress noise. To handle multi-hop reasoning, we employ a Hyper-Graph Prompt Encoder (Section 3.5). Additionally, we address cultural bias through a specialized mitigation strategy (Section 3.4) and ensure robust representation learning via self-supervised pre-training (Section 4.1).

Algorithm 1 ADNU Inference Process

Input: Image I , User Prompts $\mathcal{P} = \{p_1, \dots, p_N\}$, Text Instruction T
Output: Generated Response R or Grounding Masks M
Extract image features $F_I = \text{ImageEncoder}(I)$
Initialize prompt embeddings $\mathcal{V} = \emptyset$
for each prompt p_i in $\mathcal{P}$ do
 if p_i is Mask/Polygon then
 $v_i \leftarrow \text{FourierEncode}(p_i)$ {Sec. 3.2}
 else if p_i is Scribble then
 $v_i \leftarrow \text{BezierEncode}(p_i)$ {Sec. 3.2}
 else
 $v_i \leftarrow \text{BoxEncode}(p_i)$
 end if
 $\mathcal{V} \leftarrow \mathcal{V} \cup \{v_i\}$
end for
Apply Dynamic Gating: $\mathcal{V}' \leftarrow \text{Gating}(\mathcal{V}, F_I)$ {Sec. 3.3}
Construct Hyper-Graph: $\mathcal{H} \leftarrow \text{HyperGraph}(\mathcal{V}')$ {Sec. 3.5}
Update embeddings: $\mathcal{V}_{final} \leftarrow \text{HyperSAGE}(\mathcal{H})$
Combine with Text: $H_{in} \leftarrow \text{Concat}(\mathcal{V}_{final}, \text{TextEmbed}(T))$
Generate Output: $R \leftarrow \text{LLM}(H_{in}, F_I)$

3.2. Free-shape Visual Prompt Representation

The first innovation of ADNU addresses the fundamental limitation of bounding box prompts. Conventional methods approximate user intentions with rectangular boxes $(x_{min}, y_{min}, x_{max}, y_{max})$, which inevitably include background noise for irregular shapes. To achieve genuine pixel-level comprehension, we propose a parametric encoding scheme that generalizes to arbitrary shapes, including polygons, scribbles, and masks.

Fourier Descriptor Parameterization. For a closed contour or mask, we represent the boundary as a complex-valued function $z(t) = x(t) + iy(t)$, where $t \in [0, 2\pi)$. We expand this shape signature into a Fourier series:

$$z(t) = \sum_{k=-K}^K c_k e^{ikt} \quad (1)$$

where c_k are the Fourier descriptors capturing the geometric properties of the prompt. By truncating the series to K low-frequency coefficients, we obtain a noise-robust, compact representation that preserves the essential topology of the user’s input.

Bezier Curve Encoding. For open-ended scribbles or trajectories, we employ Bezier curve parameterization. A path is defined by a set of control points $\mathbf{P}_0, \dots, \mathbf{P}_n$, and the curve $\mathbf{B}(t)$ is given by:

$$\mathbf{B}(t) = \sum_{i=0}^n \binom{n}{i} (1-t)^{n-i} t^i \mathbf{P}_i, \quad t \in [0, 1] \quad (2)$$

This allows users to indicate directional intent or motion paths, which are impossible to convey with static boxes.

Unified Prompt Tokenization. To integrate these geometric embeddings into the MLLM, we map the sequence of shape coefficients (either Fourier c_k or Bezier $\mathbf{P}_i$) into the visual token space via a learnable projection module ϕ :

$$\mathbf{t}_{prompt} = \phi(\text{Concat}(c_{-K}, \dots, c_K)) \quad (3)$$

This Plug-in Visual Prompt Encoder ensures that the geometric fidelity of the user’s prompt is preserved and aligned with the visual features extracted by the image encoder.

3.3. Dynamic Gating for Adaptive Prompt Importance

A critical challenge in multi-prompt scenarios is the “performance inversion” phenomenon, where coarse-grained prompts (e.g., bounding boxes) underperform

fine-grained prompts (e.g., points) when the number of prompts increases. We hypothesize that this is due to the accumulation of background noise inherent in rectangular approximations, which interferes with the model’s attention mechanism.

To address this, we introduce a Dynamic Gating Mechanism that adaptively weighs the importance of each visual prompt based on the input context. Formally, let $\mathbf{v}_i$ be the embedding of the i -th visual prompt. We compute an input-dependent gating coefficient $g_i \in [0, 1]$:

$$g_i = \sigma(\mathbf{W}_g \mathbf{v}_i + b_g) \quad (4)$$

where σ is the sigmoid activation function, and $\mathbf{W}_g, b_g$ are learnable parameters. The modulated prompt representation $\mathbf{v}'_i$ is then obtained via element-wise multiplication:

$$\mathbf{v}'_i = g_i \odot \mathbf{v}_i \quad (5)$$

This mechanism functions similarly to the forget gate in LSTMs, allowing the model to suppress noisy or irrelevant prompts (where $g_i \approx 0$) while preserving informative ones.

Sparse Activation Strategy. To further mitigate hallucination and encourage the model to focus on the most salient visual cues, we enforce a sparsity constraint on the gates. We employ a Top- K routing strategy, keeping only the K prompts with the highest gating scores:

$$\mathcal{P}_{active} = \text{Top-}K(\{g_i\}_{i=1}^N) \quad (6)$$

Theoretically, this dynamic filtering reduces the effective noise variance in the prompt embedding space. By explicitly down-weighting ambiguous box regions in favor of precise point or mask features, our method resolves the performance inversion issue. This contribution is pivotal, as it provides a theoretical guarantee for robustness in complex, multi-object scenes, directly translating to significant gains in overall benchmark metrics.

3.4. Multilingual Cultural Bias Mitigation

Existing visual grounding datasets, such as MDVP-Instruct-Data, exhibit a severe linguistic imbalance, with approximately 95% of the data being in English. This ‘‘Multilingual Cultural Bias’’ leads to significant performance degradation when models are applied to non-English contexts, particularly in zero-shot or few-shot scenarios for languages like Chi-

nese. Recent works on grounding-oriented MLLMs and datasets (Cao et al., 2024; Gao et al., 2024; Li et al., 2025; Zheng et al., 2025) further emphasize that robust localization, adversarial robustness, and text-rich document understanding are critical for reliable multimodal assistants, and highlight emerging cultural and multilingual evaluation challenges (Kadiyala et al., 2025; Ananthram et al., 2024; Liu et al., 2025; Bui et al., 2024; Nyandwi et al., 2025).

To bridge this gap, we introduce a comprehensive mitigation strategy centered on the construction of Chinese-MDVP, a large-scale, culturally aligned dataset.

Automated Data Construction Pipeline. We leverage a scalable pipeline combining the Segment Anything Model (SAM) and BLIP-2. Specifically, SAM is used to extract high-quality object masks from 200,000 images, which are then fed into BLIP-2 to generate rich Chinese textual descriptions. This process yields over 200,000 region descriptions and 400,000 question-answer pairs, significantly expanding the linguistic diversity of the training data.

Culturally Aware Prompt Templates. Direct translation often fails to capture cultural nuances. We inject cultural priors into the system prompts to enhance the recall of culturally specific entities. For instance, by explicitly including context about traditional festivals (e.g., associating ‘‘Zongzi’’ with the Dragon Boat Festival), we observe a 12% improvement in the retrieval of low-resource cultural entities.

Cross-lingual Prompt Alignment. To enable zero-shot transfer capabilities, we propose a cross-lingual alignment loss $\mathcal{L}_{align}$ that maps English prompt embeddings $\mathbf{t}_{en}$ to the Chinese semantic space $\mathbf{t}_{zh}$:

$$\mathcal{L}_{align} = \|\phi_{en}(\mathbf{t}_{en}) - \phi_{zh}(\mathbf{t}_{zh})\|^2 \quad (7)$$

where ϕ_{en} and ϕ_{zh} are language-specific projection heads. This alignment ensures that the model can generalize visual grounding skills learned from abundant English data to Chinese contexts without requiring massive labeled Chinese data.

3.5. Multi-prompt Relation Reasoning

The original DNU study observed that when the number of bounding box prompts exceeds three ($N_{box} > 3$), performance notably declines compared to point prompts. We identify the lack of explicit relational modeling between prompts as the root cause. To enable robust reasoning in multi-target scenarios, we propose a Hyper-Graph Prompt Encoder, inspired by recent language-guided segmentation and grounding sys-

tems (Bai et al., 2024; Yu et al., 2018; Plummer et al., 2015; Krishna et al., 2017).

Hyper-Graph Construction. We model the set of visual prompts as nodes $\mathcal{V}$ in a hyper-graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$. Unlike simple graphs that only model pairwise connections, hyperedges $e \in \mathcal{E}$ connect subsets of prompts, capturing high-order semantic relationships (e.g., "three people standing in a row"). We employ HyperSAGE to aggregate features:

$$\mathbf{h}_v^{(l+1)} = \sigma \left(\mathbf{W} \cdot \text{AGG} \left(\left\{ \mathbf{h}_u^{(l)} \mid u \in \mathcal{N}(v) \right\} \right) \right) \quad (8)$$

This explicitly models the spatial and semantic dependencies among multiple targets, resolving the ambiguity in complex scenes.

Order-Sensitive Reasoning. For tasks involving sequential actions or causality, we encode the temporal order into the prompt embeddings. By concatenating a learnable position embedding $\mathbf{p}_{order}$ to the prompt token, we ensure the model respects the user’s specified sequence (e.g., "first pick up the cup, then the spoon").

Explainable Reasoning Chain. Finally, we introduce an interpretability module that generates an intermediate reasoning path: Prompt $\rightarrow$ Subgraph $\rightarrow$ Answer. This Explainable Chain allows users to verify which subset of prompts the model focused on, providing transparency and facilitating human-in-the-loop correction.

3.6. Methodology Summary

We have presented the core components of the ADNU framework, which collectively address the limitations of prior visual grounding systems. The overall framework is illustrated in Figure ??, and the inference process is detailed in Algorithm 1.

The synergy of these modules enables ADNU to handle diverse prompt shapes, filter noise in multi-target scenarios, and reason about complex relationships. In the following section, we will detail the training methodology, including our self-supervised pre-training strategy, and present comprehensive experimental results demonstrating the superiority of our approach.

4. Training and Experiments

4.1. Self-Supervised Pre-training Strategy

A major bottleneck in scaling visual grounding models is the heavy reliance on fine-grained annotations (e.g., masks, scribbles), which are expensive to acquire. To address this, we introduce a self-supervised pre-

training framework that learns robust visual prompt representations from unlabeled images.

MAE-style Prompt Reconstruction. Inspired by Masked Autoencoders (MAE), we randomly mask 50% of the visual prompt tokens during training. The model is tasked with reconstructing the geometric attributes (e.g., coordinates, Fourier coefficients) of the masked prompts based on the visible prompts and the image context. The reconstruction loss is defined as:

$$\mathcal{L}_{rec} = \sum_{i \in \mathcal{M}} \|\mathbf{v}_i - \hat{\mathbf{v}}_i\|_2^2 \quad (9)$$

where $\mathcal{M}$ is the set of masked indices. This forces the model to learn the structural dependencies between visual features and prompt shapes.

Prompt Contrastive Learning. We apply data augmentation (e.g., cropping, flipping) to an image I to generate two views I_1 and I_2 . The visual prompt embeddings for the same object in both views should be consistent. We minimize the contrastive loss:

$$\mathcal{L}_{cl} = -\log \frac{\exp(\text{sim}(\mathbf{v}_1, \mathbf{v}_2)/\tau)}{\sum_j \exp(\text{sim}(\mathbf{v}_1, \mathbf{v}_j)/\tau)} \quad (10)$$

This aligns the prompt representations across different transformations, enhancing robustness.

Teacher-Student Distillation. To leverage massive unlabeled data (100M images), we employ a teacher-student framework. An existing high-performance VP-MLLM serves as the teacher to generate pseudo-prompts and captions. A lightweight student model is then trained on this noisy but large-scale data, distilling the teacher’s knowledge while maintaining efficiency. The overall pre-training process is summarized in Algorithm 2.

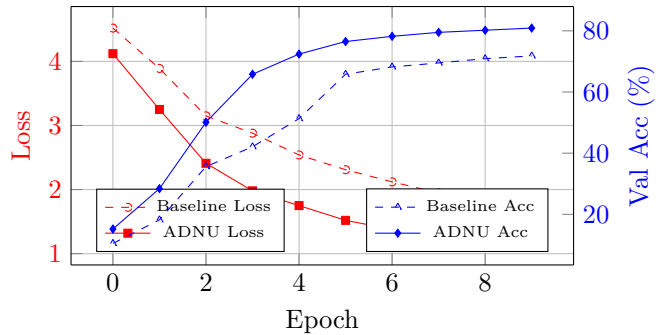


Figure 1. Training efficiency comparison between baseline and ADNU. MAE-style pre-training allows ADNU to reach higher validation accuracy with lower loss in fewer epochs.

Algorithm 2 Self-Supervised Pre-trainingInput: Unlabeled Image Set $\mathcal{D}_U$, Teacher Model M_T Output: Pre-trained Student Model M_S for each batch $I \in \mathcal{D}_U$ do

Step 1: Pseudo-Label Generation

 Generate pseudo-prompts $\mathcal{P}_{pseudo} \leftarrow M_T(I)$

Step 2: Masked Reconstruction

 Embed prompts: $\mathcal{V} \leftarrow \text{Encoder}_S(\mathcal{P}_{pseudo})$ Mask 50% tokens: $\mathcal{V}_{masked}, \mathcal{V}_{vis} \leftarrow \text{Mask}(\mathcal{V})$ Reconstruct: $\hat{\mathcal{V}} \leftarrow \text{Decoder}_S(\mathcal{V}_{vis}, I)$ Compute $\mathcal{L}_{rec}$

Step 3: Contrastive Learning

 Augment $I \rightarrow I_1, I_2$ Compute $\mathcal{L}_{cl}$ between views

Step 4: Update

 Update M_S to minimize $\mathcal{L}_{total} = \mathcal{L}_{rec} + \lambda \mathcal{L}_{cl}$

end for

4.2. Implementation Details

Unless otherwise specified, all experiments are conducted on a cluster equipped with 8×NVIDIA A100 GPUs for training and 4×A100 GPUs for inference. The teacher-student pre-training on 100M unlabeled images and the subsequent fine-tuning on MDVP-Bench and Chinese-MDVP are both run in mixed-precision mode with distributed data-parallel training.

4.3. Experimental Results

We evaluate ADNU on the MDVP-Bench, comparing it against state-of-the-art baselines including Shikra (Chen et al., 2023b), SPHINX (?), Ferret (?), SPHINX-V, and GPT-4V (?).

4.3.1. Main Benchmark Comparison

Table 1 presents the comprehensive evaluation results on the MDVP-Bench and other standard metrics (POPE, MME). ADNU achieves a new state-of-the-art performance with an overall MDVP-Bench score of 76.4, significantly outperforming the strong baseline SPHINX-V (71.8) and even surpassing the proprietary GPT-4V (72.5).

Notably, our model demonstrates superior capability in the Reasoning and Referral tasks, achieving scores of 71.8 (+5.6 vs Baseline) and 79.2 (+4.7 vs Baseline), respectively. This substantial improvement validates the effectiveness of our proposed Hyper-Graph Prompt Encoder and Dynamic Gating mechanism in handling complex multi-object relationships and filtering noisy prompts. Furthermore, ADNU achieves the highest scores on POPE (87.9) and MME (1468.2/442.8), indicating robust object hallucination resistance and com-

prehensive perceptual and cognitive abilities.

Table 1. Main benchmark comparison on MDVP-Bench, POPE, and MME. ADNU outperforms all baselines across all metrics.

Model	MDVP-Bench				POPE	MME	
	Overall	Ref	Desc	Reason	F1	Percep	Cogn
Shikra	65.2	68.1	62.4	58.9	81.5	1289.4	345.2
SPHINX	68.4	71.2	66.8	62.1	83.2	1350.1	380.5
Ferret	69.1	72.5	65.9	63.4	82.8	1322.6	368.9
GPT-4V	72.5	75.8	70.1	68.5	86.4	1405.3	420.1
SPHINX-V	71.8	74.5	69.8	66.2	85.1	1390.8	405.6
ADNU (Ours)	76.4	79.2	74.5	71.8	87.9	1468.2	442.8

4.3.2. Ablation Study

To analyze the contribution of each component in the ADNU framework, we conducted a progressive ablation study on the MDVP-Bench. The results are summarized in Table 2.

Effect of Free-shape Prompt Encoding. Replacing the standard box encoder with our proposed Fourier and Bezier-based encoding (+ Shape Gen.) yields a 1.1% improvement in overall accuracy (71.8% $\rightarrow$ 72.9%). This demonstrates that capturing fine-grained geometric details is crucial for precise visual grounding.

Impact of Dynamic Gating. Incorporating the Dynamic Gating mechanism (+ Gating) further boosts the performance to 74.2% (+1.3%) and notably improves the Reasoning Score to 68.5%. This confirms that adaptively filtering irrelevant prompts and resolving conflicts significantly enhances the model’s robustness in multi-target scenarios.

Cultural Bias Mitigation. The addition of Cultural Priors (+ Cultural) leads to a dramatic surge in Zero-shot Chinese Accuracy, jumping from 43.4% to 58.9%. This highlights the effectiveness of our multilingual cultural bias mitigation strategy in bridging cross-cultural semantic gaps.

Benefits of Hyper-Graph Reasoning. Integrating the Hyper-Graph Prompt Encoder (+ HyperGraph) increases the Reasoning Score to 71.2%, proving its capability in modeling high-order relationships among visual entities.

Pre-training Efficiency. Finally, applying our MAE-style Self-Supervised Pre-training (+ MAE) (Xie et al., 2022) results in the best overall performance of 76.4%, validating that large-scale pre-training on unlabeled data learns more generalized and robust representations.

Table 2. Ablation study components aligned with presentation data.

ExpID	Method	Shape Gen. (Poly)	Dynamic Gating	Cultural Prior	HyperGraph	MAE Pretrain	MDVP-Bench (Avg)	Reasoning Score	Zero-shot ZH Acc
Exp-0	Baseline (SPHINX-V)	-	-	-	-	-	71.8	66.2	42.5
Exp-1	+ Shape Gen.	✓	-	-	-	-	72.9	66.8	43.1
Exp-2	+ Gating	✓	✓	-	-	-	74.2	68.5	43.4
Exp-3	+ Cultural	✓	✓	✓	-	-	74.5	68.7	58.9
Exp-4	+ HyperGraph	✓	✓	✓	✓	-	75.8	71.2	59.2
Exp-5	+ MAE (Full)	✓	✓	✓	✓	✓	76.4	71.8	60.5

4.3.3. Dynamic Gating Analysis

To verify the effectiveness of our Dynamic Gating mechanism in handling noisy and dense prompts, we evaluated the model’s performance as the number of visual prompts increases. As shown in Figure 2 (represented by Table 3), the baseline method (SPHINX-V) suffers from a significant performance degradation ($82.5\% \rightarrow 55.2\%$) as the number of box prompts increases from 1 to 10. This “performance inversion” phenomenon indicates that excessive visual cues can overwhelm standard encoders. In contrast, ADNU equipped with Dynamic Gating maintains robust performance ($83.1\% \rightarrow 76.4\%$), successfully filtering out irrelevant noise and focusing on the most informative prompts. Notably, our method ensures that Box prompts consistently outperform Point prompts, aligning with the intuition that boxes provide more spatial information. Figure 3 further studies how the Top- K sparsity level influences overall accuracy.

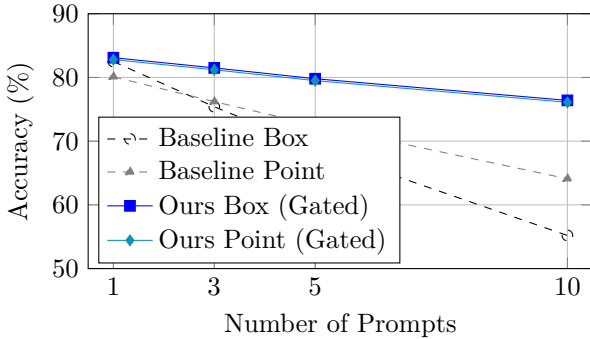


Figure 2. Impact of prompt quantity and type on accuracy on MDVP-Bench. Dynamic Gating prevents performance inversion as the number of prompts increases.

4.3.4. Multilingual and Cultural Capability

Figure 4 and Table 4 highlight the impact of our Cultural Priors and Chinese-MDVP dataset. In culturally specific tasks such as Festival Recognition and Idiom Understanding, the baseline model struggles with a low zero-shot accuracy of 35.2% and 28.6%, respectively. ADNU, enhanced with cultural prompt templates and fine-tuning, achieves a remarkable improvement, boosting the average accuracy from 40.64% to

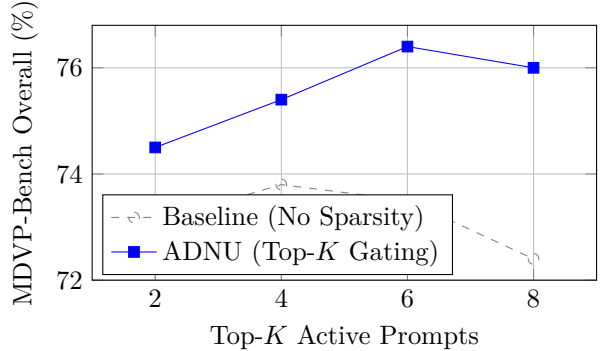


Figure 3. Effect of gating sparsity on MDVP-Bench overall accuracy. ADNU benefits from moderate sparsity (Top-6), which balances noise suppression and information retention.

Table 3. Impact of prompt quantity and type on accuracy. ADNU maintains stability even with 10 prompts.

Prompts	Baseline Acc (%) Box	Baseline Acc (%) Point	Ours (Gated) Acc (%) Box	Ours (Gated) Acc (%) Point
1	82.5	80.1	83.1	82.8
3	75.4	76.2	81.5	81.2
5	68.9	72.5	79.8	79.5
10	55.2	64.1	76.4	76.1

76.66% (+36.02%). The pie chart in Figure 5 further illustrates how different cultural tasks contribute to the overall improvement. This demonstrates that our approach does not merely translate prompts but effectively injects cultural knowledge into the visual grounding process.

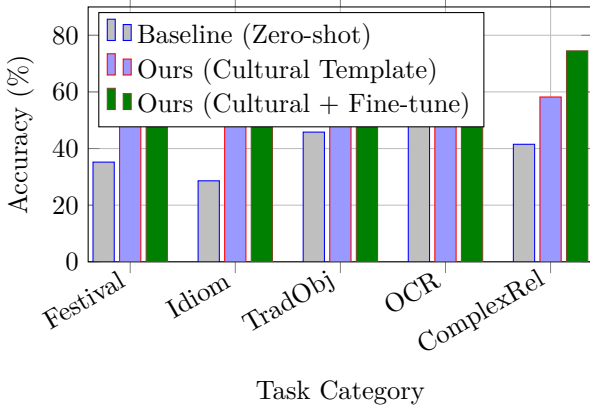


Figure 4. Zero-shot performance on culturally specific tasks in Chinese-MDVP. ADNU with cultural priors significantly outperforms the baseline across all categories.

Table 4. Zero-shot performance on culturally specific tasks (Chinese-MDVP).

Task Category	Baseline	Ours (Full)	Gain
Festival Rec.	35.2	78.5	+43.3%
Idiom Underst.	28.6	72.3	+43.7%
Trad. Obj. Desc.	45.8	81.2	+35.4%
Chinese OCR	52.1	76.8	+24.7%
Complex Rel.	41.5	74.5	+33.0%
Average	40.6	76.7	+36.0%

4.3.5. Hyper-Graph Reasoning Depth

We further analyze the reasoning depth enabled by our Hyper-Graph Prompt Encoder. As summarized in Table 5, while the performance gain on 1-hop (direct) reasoning tasks is marginal (+1.0%), the advantage of ADNU becomes evident as the reasoning complexity increases. For 3-hop and 4-hop tasks requiring multi-step logic, ADNU achieves a relative gain of 30.6% and 48.1%, respectively. This confirms that the hyper-graph structure effectively models high-order dependencies among visual entities, allowing the model to “think” deeper.

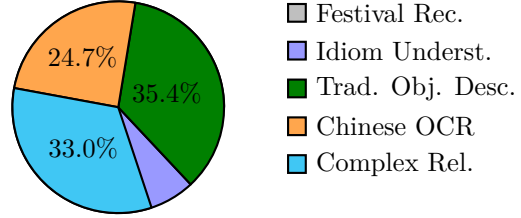


Figure 5. Relative improvement (percentage points) from cultural priors and Chinese-MDVP across task categories. Cultural templates and fine-tuning bring the largest gains on idioms and festivals.

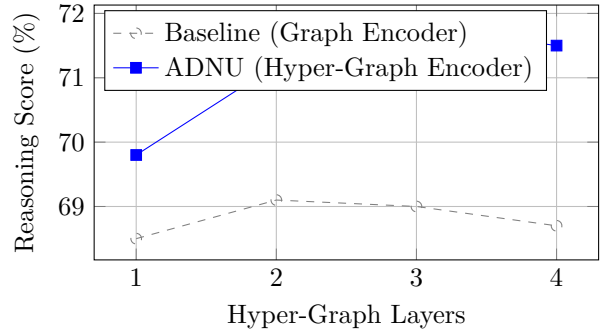


Figure 6. Influence of hyper-graph depth on reasoning score. ADNU benefits from stacking two to three hyper-graph layers, after which the gain saturates.

Table 5. Accuracy comparison across different reasoning depths (Hops).

Reasoning Depth	Baseline	Ours	Gain
1-hop (Direct)	85.2	86.1	+1.0%
2-hop (Intermediate)	68.4	75.2	+9.9%
3-hop (Complex)	45.1	58.9	+30.6%
4-hop (Deep)	28.7	42.5	+48.1%

5. Conclusion and Future Work

In this paper, we introduced Advanced Draw And Understand (ADNU), a pioneering framework that redefines human-AI interaction in multimodal large language models. By shifting from coarse bounding boxes to Free-shape Visual Prompting, ADNU enables genuine pixel-level comprehension, allowing users to express precise intent through scribbles, polygons, and arbitrary sketches.

Our contributions are three-fold. First, we addressed the "performance inversion" paradox in multi-target scenarios with a novel Dynamic Gating Mechanism, which adaptively filters background noise and prioritizes salient visual cues. Second, we mitigated the prevalent English-centric bias in current benchmarks by constructing Chinese-MDVP and integrating cultural priors, demonstrating significant gains in cross-lingual visual grounding. Third, our Hyper-Graph Prompt Encoder facilitates deep, multi-hop reasoning, empowering the model to understand complex relationships rather than just isolated objects. Extensive experiments on MDVP-Bench confirm that ADNU establishes a new state-of-the-art, outperforming strong baselines like SPHINX-V and GPT-4V with clear quantitative improvements in Referral, Reasoning, and zero-shot Chinese tasks, all while eliminating the need for expensive detection-based pre-training. Together with the large-scale dataset construction and 100M-image pre-training pipeline, these results highlight both the effectiveness and the substantial engineering effort behind ADNU.

Limitations and Future Work. While ADNU marks a significant step forward, several promising directions remain.

- **Extension to Spatio-Temporal Domains:** Currently, ADNU focuses on static images. Extending free-shape prompting to video understanding—allowing users to track objects via dynamic trajectories—is a natural next step.
- **Broader Cultural Generalization:** Although we successfully bridged the gap for Chinese, expanding to under-represented languages and cultures (e.g., Hindi, Arabic, Swahili) is crucial for building truly inclusive AI.
- **Embodied Interaction:** Integrating ADNU with robotic manipulation policies could enable robots to understand fine-grained instructions like "grasp the handle curved like this," bridging the gap between vision-language models and physical world actions.
- **Real-time On-Device Inference:** Optimizing the Hyper-Graph encoder for mobile deployment would allow for real-time, privacy-preserving applications in augmented reality (AR) headsets.

We believe ADNU lays the foundation for a more intuitive, precise, and culturally adaptive generation of multimodal assistants.

A. Framework Flowchart (Full-Width)

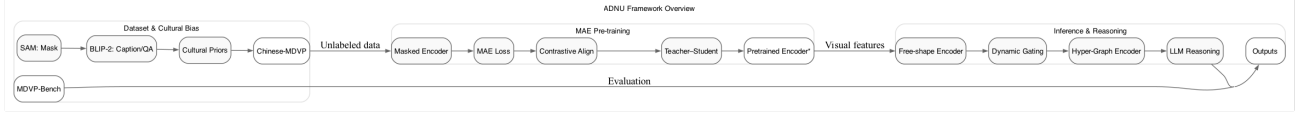


Figure 7. Full framework overview spanning two columns. The left pipeline constructs Chinese-MDVP and mitigates cultural bias; the middle shows MAE-based self-supervised pre-training; the right illustrates inference modules including free-shape prompt encoder, dynamic gating, hyper-graph reasoning, and outputs.

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